Credit Card Fraud Detection Analysis

BIL 476 / 573 — Data Mining Course Project, Spring 2026

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***Abstract*—Credit card fraud detection is a difficult problem mainly because fraudulent transactions are so rare. In a typical dataset, fraud makes up well under 1% of all records, which causes most classifiers to simply ignore it. This project uses a publicly available dataset of anonymized European card transactions to compare four machine learning configurations under these conditions: two supervised Random Forest variants (with and without SMOTE-based oversampling), and two unsupervised methods, Isolation Forest and Local Outlier Factor. Precision, recall, and the area under the precision-recall curve were used as the main evaluation metrics, since overall accuracy is essentially meaningless here. Surprisingly, the unbalanced Random Forest achieved a higher F1-Score (0.880) than its SMOTE-augmented counterpart (0.558), suggesting that the PCA-transformed features provide sufficient separability for supervised learning without explicit balancing. Both supervised approaches substantially outperformed the unsupervised methods. Isolation Forest caught only about a third of fraud instances, and Local Outlier Factor performed no better than random guessing. The main takeaway is that labeled data and supervised boundary learning make a substantial difference over standard anomaly detection, regardless of whether explicit class balancing is applied.**

***Index Terms*—data mining, credit card fraud, imbalanced data, anomaly detection, SMOTE**

I. Introduction

Credit card fraud causes significant financial losses worldwide each year. Building automated systems to catch it is important, but the core difficulty is that fraud is extremely rare in real transaction data — often less than 0.2% of all records. When a classifier is trained on such skewed data, it tends to predict everything as normal and still achieve very high accuracy. This makes standard accuracy essentially useless as a performance measure.

The goal of this project is to compare different machine learning algorithms on this type of imbalanced problem and see which ones actually work in practice. Three methods are evaluated: a supervised approach using Random Forest with SMOTE-based oversampling, and two unsupervised anomaly detection algorithms, Isolation Forest and Local Outlier Factor (LOF). The core components of the approach are as follows:

- **Data Balancing:** SMOTE is applied to the training set to give the minority class a fair representation before the supervised model is trained.
- **Model Comparison:** Random Forest is compared against Isolation Forest and LOF, representing supervised and unsupervised paradigms respectively. An unbalanced Random Forest baseline is additionally included to isolate the contribution of SMOTE.
- **Performance Evaluation:** Precision, Recall, and PR-AUC are used instead of accuracy, since these metrics are actually informative when one class is extremely rare.

Section II reviews the relevant literature. Section III describes the dataset. Section IV explains the preprocessing and modeling steps. Section V presents the results, Section VI interprets them, and Section VII concludes with future directions.

II. Related Work

Fraud detection on imbalanced datasets has been studied extensively. Most prior work falls into two broad areas: handling class imbalance, and applying specific machine learning algorithms.

- **Handling Imbalanced Data:** Chawla et al. [1] introduced SMOTE, which generates synthetic minority class samples by interpolating between existing instances rather than duplicating them. Their experiments showed improved decision boundaries on standard UCI benchmarks. This project applies their technique to high-dimensional, PCA-transformed financial data, which is a different setting from what they originally tested.
- **Supervised Approaches:** Dal Pozzolo et al. [2] studied Random Forest specifically in the context of credit card fraud and found it to be robust against class imbalance and concept drift. Their work mostly relied on traditional sampling strategies; here, SMOTE is integrated and results are evaluated using PR-AUC rather than ROC metrics, which better reflects real-world detection capability.
- **Unsupervised Anomaly Detection:** Liu et al. [3] proposed Isolation Forest, which isolates anomalies by randomly partitioning the feature space — anomalies tend to be separated with fewer splits. Breunig et al. [4] proposed LOF, a density-based method that flags points whose local neighborhood density is much lower than

their neighbors'. Both are commonly applied to fraud detection problems.

- **Gap and Contribution:** Many papers test supervised and unsupervised methods separately, or on datasets where fraud patterns are more distinguishable. Direct comparisons on heavily anonymized, PCA-transformed financial data are less common. This project benchmarks all three approaches on the same dataset under the same conditions, which makes the limitations of unsupervised methods more apparent.

III. Dataset Description

A. Data Source

The dataset utilized in this study is the well-known "Credit Card Fraud Detection" dataset, publicly available on the Kaggle platform. It contains transactions made by European cardholders over a span of two days in September 2013. This dataset is highly appropriate for the chosen problem of financial anomaly detection for several reasons:

- **Real-World Representation:** It reflects the true nature of financial transaction data, including extreme class imbalance and high dimensionality.
- **Algorithmic Challenge:** The extreme scarcity of fraudulent events provides an ideal testbed for evaluating the robustness of algorithms under severe skewness.
- **Standardized Benchmark:** As a widely recognized dataset in the data mining community, it allows the findings of this study to be contextualized within the broader academic literature.

B. Data Characteristics

The dataset consists of continuous numerical features. Due to confidentiality agreements, the original background information and feature names are not provided.

Table I provides a comprehensive summary of the structural properties of the dataset.

TABLE I
Dataset Summary

Property	Value
Number of Records	284,807
Number of Attributes	31 (30 input features + 1 target)
Attribute Types	Numeric (Continuous)
Target Variable	Class (0: Normal, 1: Fraud)
Missing Values	0 (Data is completely populated)
Class Distribution	99.827% Normal / 0.173% Fraud
Source URL	https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

A deeper exploratory data analysis reveals crucial details regarding the non-transformed attributes, specifically *Time* and *Amount*:

- **Feature 'Time':** Represents the seconds elapsed between each transaction and the first transaction in the dataset. It ranges from 0 to 172,792 seconds (exactly 48 hours).
- **Feature 'Amount':** Represents the transaction value. Descriptive statistics show a mean of 88.34 and a median of 22.00, with a maximum value of 25,691.16. The massive discrepancy between the mean and median indicates a heavily right-skewed distribution containing significant outliers.
- **Features 'V1' to 'V28':** These are the principal components obtained via Principal Component Analysis (PCA). They are already scaled and centered around zero as a result of the transformation.

A class-separated analysis of the scaled transaction amounts reveals a notable pattern: fraudulent transactions show a median scaled amount of -0.18, compared to 0.00 for normal transactions, indicating that fraud cases tend to cluster around lower monetary values. The PCA-transformed features V1–V28 exhibit near-zero means (confirming correct centering), while their standard deviations decrease monotonically from V1 (1.959) to V28 (0.330), reflecting the variance-ordered nature of PCA decomposition.

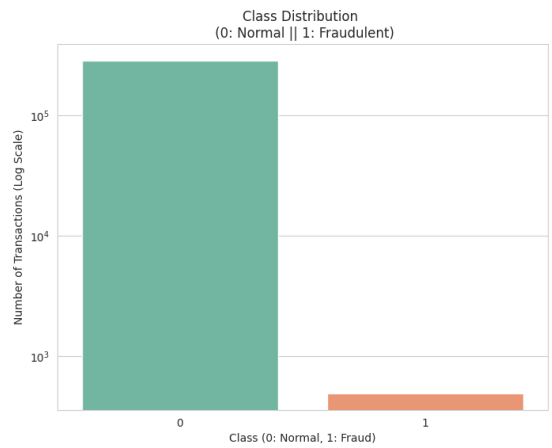

Fig. 1. Class Distribution demonstrating extreme imbalance (0: Normal, 1: Fraudulent).

Fig. 2. Transaction Amount Distribution: Normal vs. Fraud classes (extreme outliers removed for visual clarity).

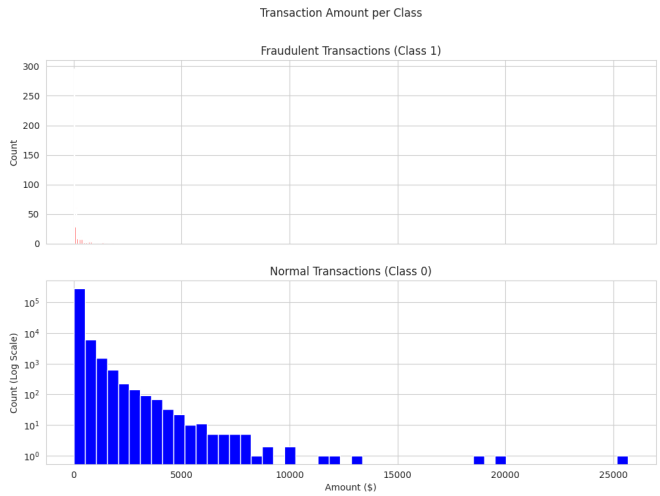

Fig. 3. Distribution of transaction amounts per class. The log-scale for normal transactions highlights the extreme frequency difference.

C. Ethical Considerations

Working with financial transaction data inherently raises critical ethical and privacy concerns. In this study, these considerations are systematically addressed in two primary domains:

- **Data Privacy and Anonymity:** To protect user confidentiality, the data source has stripped all personal identifiers (such as account numbers, names, and locations). The application of PCA transformation on features V1 through V28 ensures that sensitive financial behaviors cannot be reverse-engineered or traced back to individual cardholders. Therefore, privacy is rigorously maintained.
- **Algorithmic Bias and Fairness:** The extreme class imbalance (99.827% normal transactions) introduces a significant risk of algorithmic bias. If trained directly on this data, a machine learning model would inherently favor the majority class, systematically ignoring the minority (fraudulent) transactions. To ensure algorithmic fairness and prevent the model from developing a blind spot toward illicit activities, data balancing techniques (specifically SMOTE) are implemented in the methodology phase. This ensures the predictive model evaluates anomalies objectively rather than being biased by the sheer volume of normal data.

IV. Methodology

A. Data Preprocessing

Data preprocessing is conducted to ensure the stability and predictive power of the models. Several critical steps are performed:

- **Handling Missing Values:** Preliminary analysis confirmed that the dataset contains no missing values; therefore, imputation techniques were not required.
- **Normalization / Standardization:** Since the features Time and Amount contain significant outliers (as evidenced by the large discrepancy between the mean and median of Amount), RobustScaler is applied instead of StandardScaler. RobustScaler scales features using the median and the interquartile range, making it robust to outliers and more suitable for this dataset.
- **Dimensionality and Feature Relevance:** Although the V1-V28 features are already reduced via PCA, their individual relevance to the target variable is assessed using a correlation heatmap (Fig. 4). Strong negative correlations in features like V17 and V14, and positive correlations in V11, indicate high discriminative power.
- **Handling Class Imbalance (SMOTE):** To prevent the models from becoming biased toward the majority class, the **Synthetic Minority Over-sampling Technique (SMOTE)** is applied specifically to the training set. This ensures the minority class (fraud) is represented equally without the risks of simple over-sampling, such as overfitting.

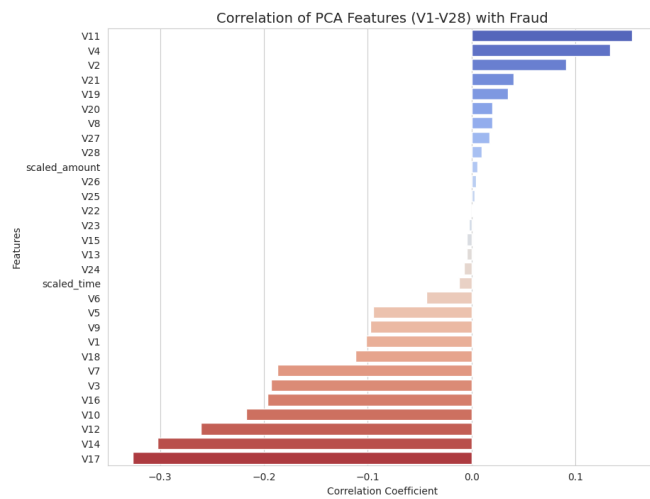

Fig. 4. Correlation of PCA features with the target Class variable.

B. Algorithm Selection and Description

Three distinct machine learning philosophies are compared to identify the most effective approach for financial anomaly detection:

- **Random Forest (Supervised):** An ensemble learning method that constructs multiple decision trees during training. It is chosen for its robustness against high-dimensional data and its ability to capture non-linear relationships through bagging.
- **Isolation Forest (Unsupervised):** An anomaly detection algorithm that isolates observations by randomly selecting a feature and a split value. It is theoretically appropriate for fraud detection because it assumes anomalies are "few and different," making them easier to isolate in shorter tree branches.
- **Local Outlier Factor (Unsupervised):** A density-based algorithm that identifies anomalies by comparing the local density of a point to its neighbors. It is selected to

evaluate whether fraud instances can be detected as low-density deviations in the feature space.

C. Experimental Setup

The experimental workflow is designed to maintain the integrity of the evaluation process:

- **Data Splitting and Validation:** The dataset is partitioned into a training set (80%) and a hold-out test set (20%) using stratified sampling to preserve the original 0.17% fraud ratio. To validate the stability of the Random Forest model and guard against overfitting on the SMOTE-balanced training data, a 5-fold Stratified Cross-Validation was performed on the SMOTE-balanced training data to verify model stability, yielding a mean F_1-Score of 0.990. Note that this score reflects performance on synthetic data and serves as a consistency check rather than an unbiased generalization estimate. The final performance evaluation is conducted on the held-out, unbalanced test set to reflect real-world deployment conditions.
- **Balancing Strategy:** SMOTE is applied only to the training data. The class distribution before and after SMOTE is visualized in Fig. 5. The test set remains unbalanced to simulate a real-world environment.
- **Software Environment:** The project is implemented in Python 3.x using the *scikit-learn* library for modeling and *imbalanced-learn* for SMOTE. Data visualization is performed using *seaborn* and *matplotlib*.
- **Hyperparameter Configuration:**
  - **Random Forest:** n_estimators=50, max_depth=10.
  - **Isolation Forest:** n_estimators=100, contamination=0.0017.
  - **LOF:** n_neighbors=20.

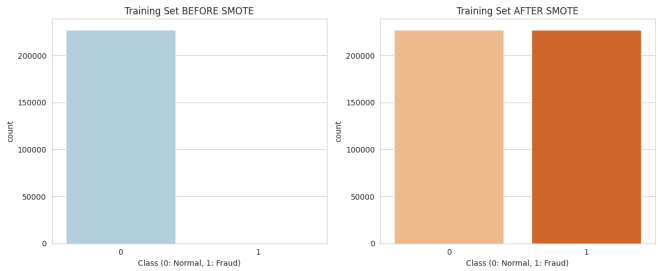

Fig. 5. Class distribution in the training set before and after SMOTE application.

V. Results

A. Evaluation Metrics

Standard classification accuracy is highly misleading for this task because a native classifier predicting all transactions as legitimate would achieve 99.83% accuracy while failing to detect any fraud. To prioritize the detection of rare illicit activities, the following robust metrics are utilized:

- **Recall (Sensitivity):** Critical for minimizing False Negatives, measuring the percentage of actual fraudulent transactions correctly identified.
- **Precision:** Essential for minimizing False Positives, ensuring that normal customers are not unnecessary inconvenienced by false alarms.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced performance metric for the minority class.

While these provide point estimations, the overall classifier capability is assessed through *Receiver Operating Characteristic (ROC)* and *Precision-Recall (PR)* area-under-the-curve metrics.

B. Results Summary

Table II presents the complete comparison including a baseline Random Forest trained without any class balancing. A notable finding is that the unbalanced Random Forest achieves a considerably higher F1-Score (0.880) and Precision (0.942) than its SMOTE-augmented counterpart (F1: 0.558, Precision: 0.414), while its Recall is only marginally lower (0.827 vs. 0.857, a 3.7% difference). This result suggests that the PCA-transformed feature space of this particular dataset provides sufficient discriminative power for Random Forest to learn fraud boundaries without explicit balancing. Both supervised models substantially outperform the unsupervised methods.

TABLE II
Model Comparison Summary Table

Model	Precision	Recall	F1-Score	PR-AUC
Random Forest (No SMOTE)	0.942	0.827	0.880	0.863
Random Forest (SMOTE)	0.414	0.857	0.558	0.801
Isolation Forest	0.297	0.337	0.316	0.162
Local Outlier Factor	0.005	0.143	0.010	0.003

C. Visualizations

The technical evidence for our findings is presented through comparative curves and individual error analysis. Fig. 6 and Fig. 7 illustrate the performance trade-offs for all four model configurations, including the near-baseline performance of LOF on the precision-recall space (PR-AUC = 0.003), while the individual confusion matrices (Fig. 8, 9, 10) provide the exact classification counts.

D. Comparison of Approaches

The comparative analysis yields several critical implications regarding the application of data mining techniques to financial anomaly detection:

- **The Superiority of Supervised Learning:** Both supervised Random Forest variants substantially outperformed all unsupervised methods. The SMOTE-augmented model achieved the highest Recall (0.857), while the unbalanced baseline achieved the highest F1-Score (0.880) and PR-AUC (0.863). In both cases, supervised boundary learning proved far superior to density-based or isolation-based anomaly detection. This proves

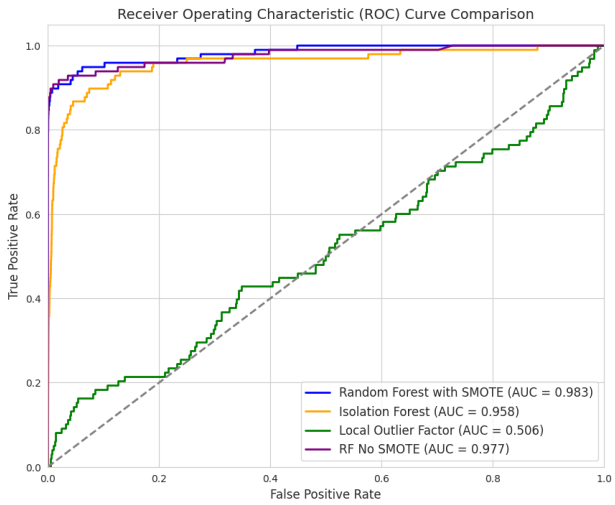

Fig. 6. Receiver Operating Characteristic (ROC) Curve Comparison.

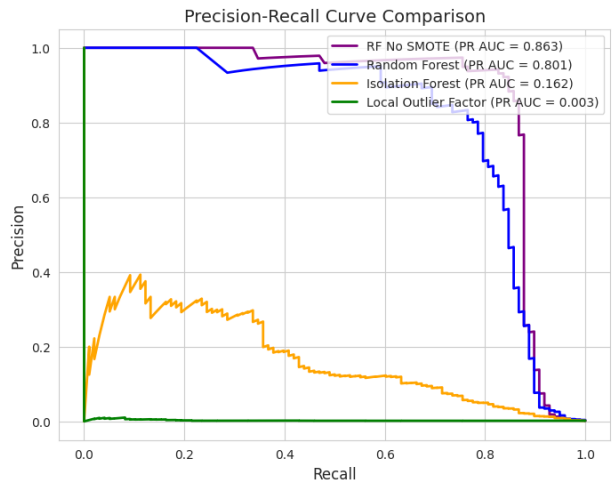

Fig. 7. Precision-Recall Curve Comparison.

that specifically learning the boundaries of illicit behavior is superior to assuming anomalies are simply sparse outliers.

- **The Recall-Precision Trade-off:** The moderate Precision (0.414) in the SMOTE-augmented Random Forest is a standard trade-off in high-stakes fraud detection, where the cost of a missed fraud (False Negative) is exponentially higher than the administrative cost of reviewing a false alarm (False Positive).
- **Metric Specificity:** The high ROC AUC of Isolation Forest (0.958) contrasted with its low Recall (0.337) and PR AUC (0.162) proves that ROC analysis can give an overly optimistic view in imbalanced contexts. PR-AUC is the more reliable descriptor of actual detection capability in real-world scenarios. Similarly, the Local Outlier Factor achieved a PR-AUC of only 0.003 , effectively no better than random chance, confirming its complete inability to separate fraud from normal transactions in this high-dimensional space.

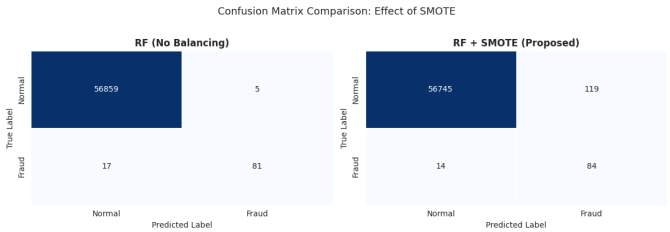

Fig. 8. Confusion matrix comparison: RF without balancing (left) vs. RF + SMOTE (right). The unbalanced model achieves higher precision but misses more fraud cases (FN: 17 vs. 14).

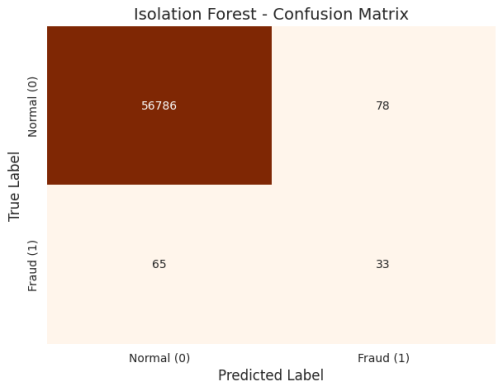

Fig. 9. Isolation Forest CM

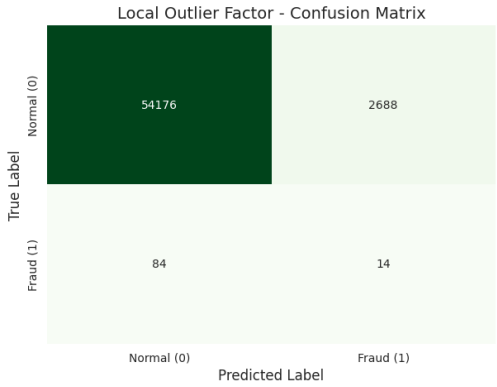

Fig. 10. LOF CM

VI. Discussion

The experimental results provide several insights into the effectiveness of machine learning paradigms for financial fraud detection. A critical analysis of these findings reveals the following observations:

- **Superiority of Supervised Learning:** The SMOTE-augmented Random Forest achieved the highest Recall (0.857), while the unbalanced baseline achieved the highest F1-Score (0.880) and PR-AUC (0.863). In the context of this dataset, both supervised variants substantially outperformed all unsupervised methods. This success is

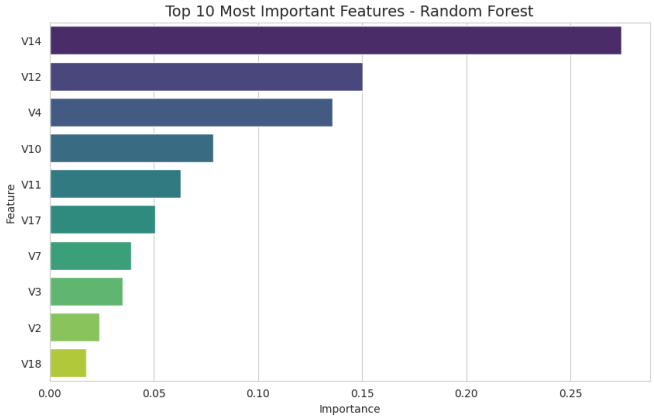

Fig. 11. Top-10 Feature Importances for Random Forest: V14 and V12 are the most discriminative, followed by V4 and V10.

attributed to the ensemble nature of Random Forest, which effectively captures non-linear relationships in the PCA-transformed feature space.

Unexpectedly, the unbalanced Random Forest outperformed the SMOTE-augmented model on F1-Score and PR-AUC. This can be attributed to the strong discriminative power of the PCA-transformed features – particularly V14, V12, and V4 – which allow the model to distinguish fraud from normal transactions even with only 394 training examples of the minority class. SMOTE marginally improved Recall (+3.7%) but caused a severe Precision drop ($0.942 \rightarrow 0.414$), indicating that the synthetic fraud samples generated by interpolation pushed the decision boundary outward, flagging many legitimate transactions as fraudulent. This finding is consistent with reports in the literature that SMOTE provides diminishing returns on well-separated datasets, and that its benefit is most pronounced when the minority class overlaps heavily with the majority class in feature space.

- **Feature Importance Alignment:** A critical observation from the Feature Importance analysis (Fig. 11) is that features V14, V12, V4, and V10 are the primary drivers of the Random Forest model's decisions. This result is highly consistent with the earlier correlation analysis, where these features demonstrated the strongest relationships with fraudulent activity. This alignment confirms that our model successfully prioritized the most relevant financial indicators, despite the anonymized nature of the PCA components.
- **The ROC vs. PR Paradox:** A surprising finding was the discrepancy in Isolation Forest's performance metrics. While it achieved an impressive ROC-AUC of 0.958, its PR-AUC collapsed to 0.162. This demonstrates a critical limitation in imbalanced data mining: ROC-AUC can be overly optimistic as it accounts for the correct classification of the majority class. In contrast, the PR curve highlights the model's struggle to maintain precision when fraud is rare, confirming that unsupervised isolation is insufficient for sophisticated fraud patterns.
- **Impact of Data Imbalance:** The extreme skewness (0.17% fraud) was the primary driver of the results. Without SMOTE, the Random Forest still demonstrated strong performance on this dataset due to the high discriminative power of the PCA-engineered features, achieving a Recall of 0.827 and F1 of 0.880. This contrasts with the expected behaviour on less separable datasets, where unbalanced training typically causes classifiers to neglect the minority class entirely. The complete failure of the Local Outlier Factor (F1-Score: 0.01) further proves that density-based assumptions do not hold in high-dimensional PCA-transformed spaces where fraudulent transactions are designed to mimic normal behavior.
- **Limitations:** A significant limitation of this study is the use of synthetic data generation. While SMOTE improves recall, it may introduce noise by creating unrealistic transactions that do not exist in real-world scenarios. Furthermore, the PCA-transformed nature of the features prevented domain-specific feature engineering, such as analyzing geographic locations or merchant categories, which could potentially improve precision.
- **Ethical Implications and Fairness:** From an ethical perspective, the use of PCA ensures cardholder privacy, satisfying data protection standards. However, the moderate precision (0.414) of the SMOTE-augmented model, which achieves the highest Recall, introduces "customer friction." Flagging legitimate transactions as fraud can cause inconvenience and financial distress for users. There is an inherent trade-off between the security of the financial institution (high recall) and the seamless experience of the customer (high precision). Fairness dictates that models must be continuously tuned to minimize this friction without compromising safety.

VII. Conclusion and Future Work

This study conducted a comprehensive comparative analysis of supervised and unsupervised machine learning algorithms for credit card fraud detection under conditions of extreme class imbalance. The key findings and achievements of the project are summarized below:

- **Key Achievements:** The project demonstrated that supervised Random Forest models consistently outperform unsupervised anomaly detection methods on this task. Notably, the unbalanced baseline achieved the highest F1-Score (0.880), while SMOTE improved Recall at the cost of precision, yielding complementary trade-offs depending on deployment priorities. The final model achieved a high Recall of 0.857, effectively capturing the vast majority of illicit activities that standard algorithms would otherwise overlook.
- **Core Takeaways:** The most significant takeaway is the critical importance of selecting appropriate evaluation metrics. While unsupervised models like Isolation Forest showed misleadingly high ROC-AUC scores, their failure in Precision-Recall (PR) analysis revealed their practical

limitations. It is concluded that supervised learning is far superior to density-based or isolation-based anomaly detection for this domain, regardless of whether class balancing is applied for this specific financial domain.

- **Process Improvements:** If more time were available, a more extensive hyperparameter optimization phase using GridSearch or Bayesian Optimization could have been implemented to further refine the precision-recall trade-off. Additionally, exploring hybrid techniques that combine the strengths of both supervised and unsupervised models could potentially reduce the false positive rate.
- **Future Directions:** Concrete directions for future research include:
  - **Deep Learning:** Implementing Autoencoders [5] or Generative Adversarial Networks (GANs) to better model the underlying distribution of normal transactions.
  - **Cost-Sensitive Learning:** Integrating a cost matrix into the loss function to explicitly penalize missing a fraud (False Negative) more heavily than a false alarm.
  - **Concept Drift Analysis:** Evaluating the model's performance on time-series data to account for how fraud patterns evolve over time in a production environment.

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APPENDIX

The complete source code, including data exploration notebooks, model training scripts, and visualization utilities, is publicly available on GitHub at the following repository: https://github.com/emiryucedag/Credit-Card-Fraud-Detection-Analysis

```python
from sklearn.preprocessing import RobustScaler
from imblearn.over_sampling import SMOTE
from sklearn.ensemble import RandomForestClassifier

rob_scaler = RobustScaler()
df['scaled_amount'] = rob_scaler.fit_transform(df['Amount'].values.reshape(-1,1))
df['scaled_time'] = rob_scaler.fit_transform(df['Time'].values.reshape(-1,1))

sm = SMOTE(random_state=42)
X_train_smote, y_train_smote = sm.fit_resample(X_train, y_train)

rf_model = RandomForestClassifier(n_estimators=50, max_depth=10, random_state=42, n_jobs=-1)
rf_model.fit(X_train_smote, y_train_smote)
```

Listing 1. Implementation of Scaling, SMOTE, and Random Forest

In this section, supplementary materials that support the findings in Section V are provided. Table III presents the per-class precision and recall values for the Random Forest model, highlighting the performance on the minority class.

TABLE III
DETAILED CLASSIFICATION REPORT FOR RANDOM FOREST (SMOTE)

Class	Precision	Recall	F1-Score	Support
Normal (0)	1.00	1.00	1.00	56,864
Fraud (1)	0.41	0.86	0.56	98
Average	1.00	1.00	1.00	56,962

Note: The support values reflect the 20% test split of the original 284,807 transactions. Any additional EDA plots or hyperparameter tuning logs can also be accessed via the code repository mentioned in Appendix A.